

Somaiya Vidyavihar University
K J Somaiya School of Engineering

Batch: A2 Roll No.: 16010324044

Experiment / assignment / tutorial No. 06

Grade: AA / AB / BB / BC / CC / CD / DD

Signature of the Staff In-charge with date

Title: Line Coding

Objectives:

1. To understand different line coding techniques of base band signal
2. To develop a logic and generate waveforms of the given line coding techniques

OUTCOME: analyze distortion less baseband transmission and reception of signal along with waveform coding techniques

Problem statement:

A. For the given binary sequence $b = \{1, 0, 1, 0, 1, 1\}$; sketch the waveforms representing the sequence b using the following line codes:

- Unipolar NRZ
- Polar NRZ
- Bipolar NRZ
- Manchester

Assume unit pulse amplitude and use binary data rate $R_b = 1\text{ kbps}$. Write a program using MATLAB or any other open source software for the above line codes.

B. Determine and sketch the power spectral density (PSD) functions corresponding to the above line codes.

Hints: Use $R_b = 1\text{ kbps}$: Let $f_1 > 0$ be the location of the first spectral null in the PSD function. If the transmission bandwidth B_T of a line code is determined by f_1 , determine B_T for the line codes in question 1 as a function of R_b :

Let f_{p1} : first spectral peak; f_{n1} : first spectral null; f_{p2} : second spectral peak; f_{n2} : second spectral null; such that all $f_{p/n} > 0$

Record your observations in Table:

$R_b =$	F_{p1}	F_{N1}	F_{p2}	F_{N2}	B_T
Unipolar NRZ	1 kHz	1.43 kHz	2.46 kHz	2 kHz	1 kHz
Polar NRZ	1 kHz	1.43 kHz	2.46 kHz	2 kHz	1 kHz
Bipolar NRZ	1 kHz	0.37 kHz	1 kHz	1.47 kHz	1 kHz
Manchester	1 kHz	0.74 kHz	2 kHz	2.93 kHz	2 kHz

Expected output:

- Waveforms of all line coding techniques
- Graph of magnitude response of each waveform
- Bandwidth required for transmitting each line code of the baseband waveform.

Code:

```

clc;
clear;
close all;
b = [1 0 1 0 1 1];
Rb = 1000;
N = 100;
Tb = 1/Rb;
%% Line Coding
% Bipolar NRZ / AMI
a = 2*mod(cumsum(b)-1,2)-1;
a(b==0) = 0;
% Manchester
m = reshape([2*b-1; 1-2*b],1,[]);
% Generate waveforms
X = [repelem(b,N);
     repelem(2*b-1,N);
     repelem(a,N);
     repelem(m,N/2)];
name = {'Unipolar NRZ','Polar NRZ','Bipolar NRZ','Manchester'};
%% =====
%          WAVEFORMS
% =====
figure;
subplot(5,1,1);
plot(repelem(b,N),'LineWidth',1.2);
grid on;
title('Input Binary Signal');
ylabel('Binary');
ylim([-0.5 1.5]);
for k = 1:4
    subplot(5,1,k+1);
    plot(X(k,:),'LineWidth',1.2);
    grid on;
    title(name{k});
    ylabel('Amplitude');
    ylim([-1.5 1.5]);
end
sgtitle('Input and Line Coding - 101011');
```

```

%% =====
% THEORETICAL PSD
% =====
f = linspace(0,5*Rb,3000);
z = f*Tb;
P = [.25*sinc(z).^2;
     sinc(z).^2;
     .5*sinc(z).^2.*sin(pi*z).^2;
     sinc(z/2).^2.*sin(pi*z/2).^2];
figure;
for k = 1:4
    subplot(4,1,k);
    plot(f/1000,P(k,:),'LineWidth',1.2);
    grid on;
    title(['PSD - ' name{k}]);
    xlabel('Frequency (kHz)');
    ylabel('Power');
    xlim([0 5]);
end
sgtitle('Theoretical Power Spectral Density');
%% =====
% MAGNITUDE SPECTRUM
% =====
% FFT
Y = abs(fft(X,[],2))/size(X,2);
% Keep positive frequencies
Y = Y(:,1:size(X,2)/2+1);
% Frequency axis
F = (0:size(Y,2)-1)*Rb*N/size(X,2);
figure;
for k = 1:4
    subplot(4,1,k);
    plot(F/1000,Y(k,:),'LineWidth',1.2);
    grid on;
    title(['Magnitude Spectrum - ' name{k}]);
    xlabel('Frequency (kHz)');
    ylabel('|X(f)|');
    xlim([0 5]);
end
sgtitle('Magnitude Spectrum of Line Codes');
%% =====
% BANDWIDTH
% =====
BW = [Rb Rb Rb 2*Rb];
fprintf('\nBandwidth (First Spectral Null):\n');
fprintf(['Unipolar NRZ = %.1f kHz\n' ...
        'Polar NRZ = %.1f kHz\n' ...
        'Bipolar NRZ = %.1f kHz\n' ...
        'Manchester = %.1f kHz\n'], BW/1000);

```

Output:

Figure 1 × Figure 2 × Figure 3 ×

Input and Line Coding - 101011

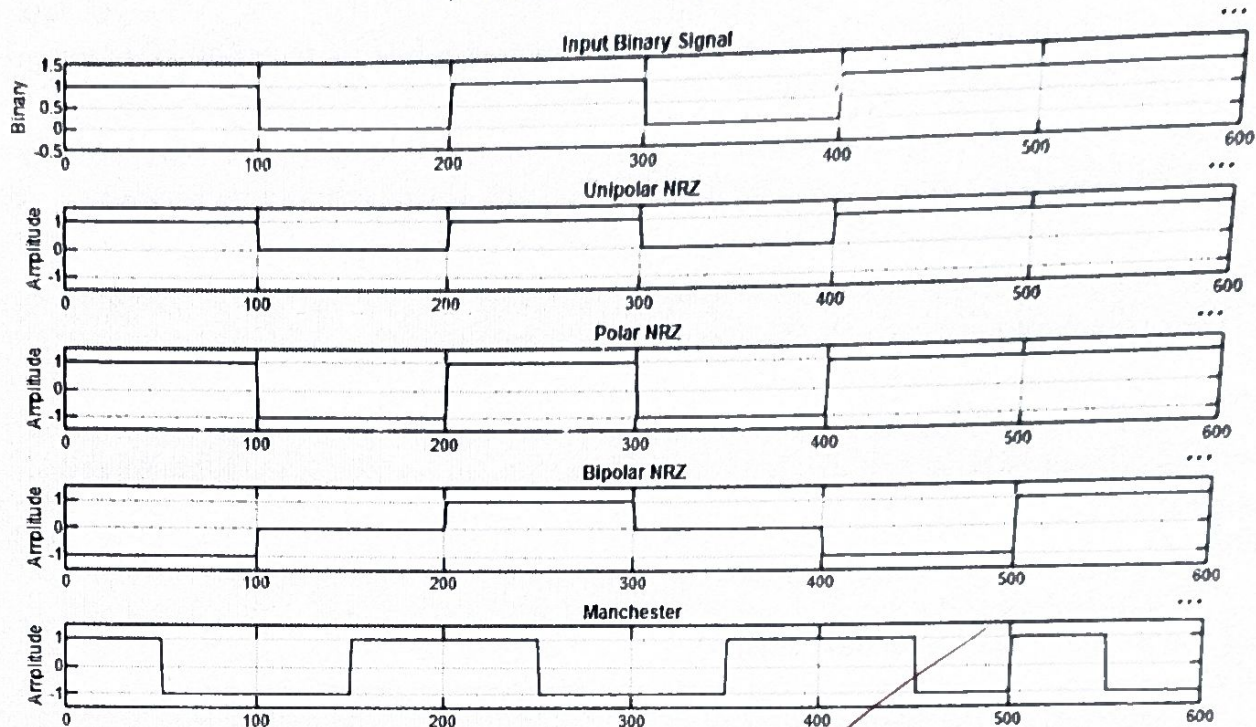
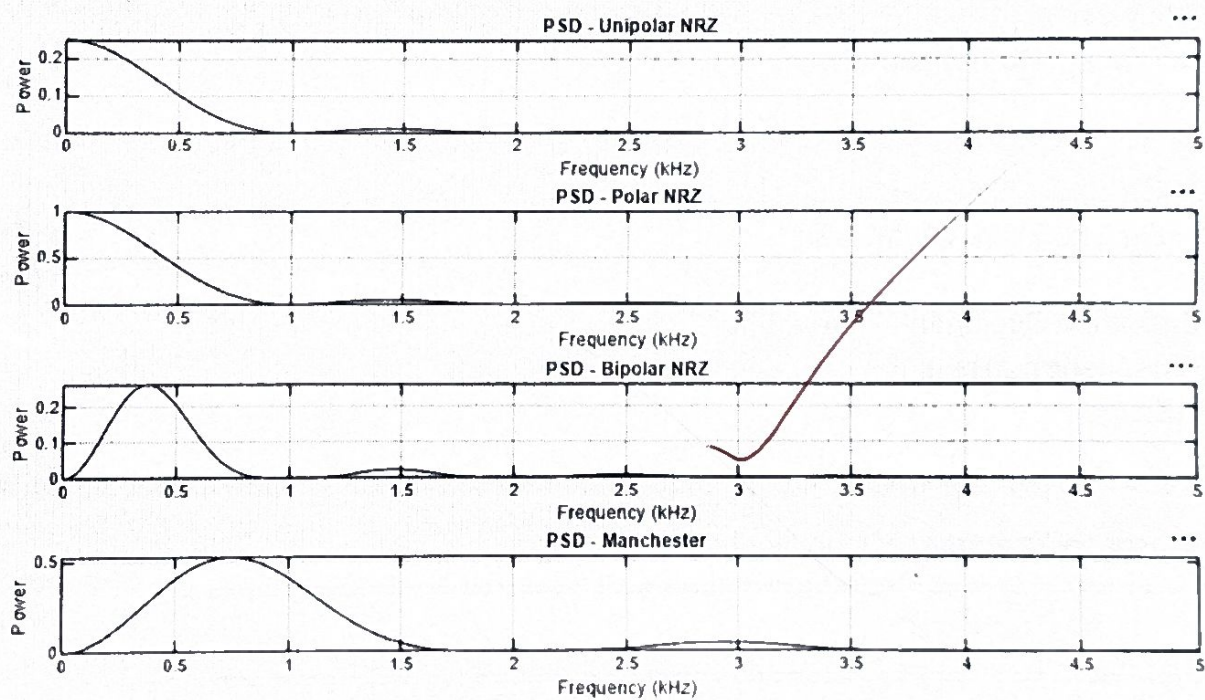
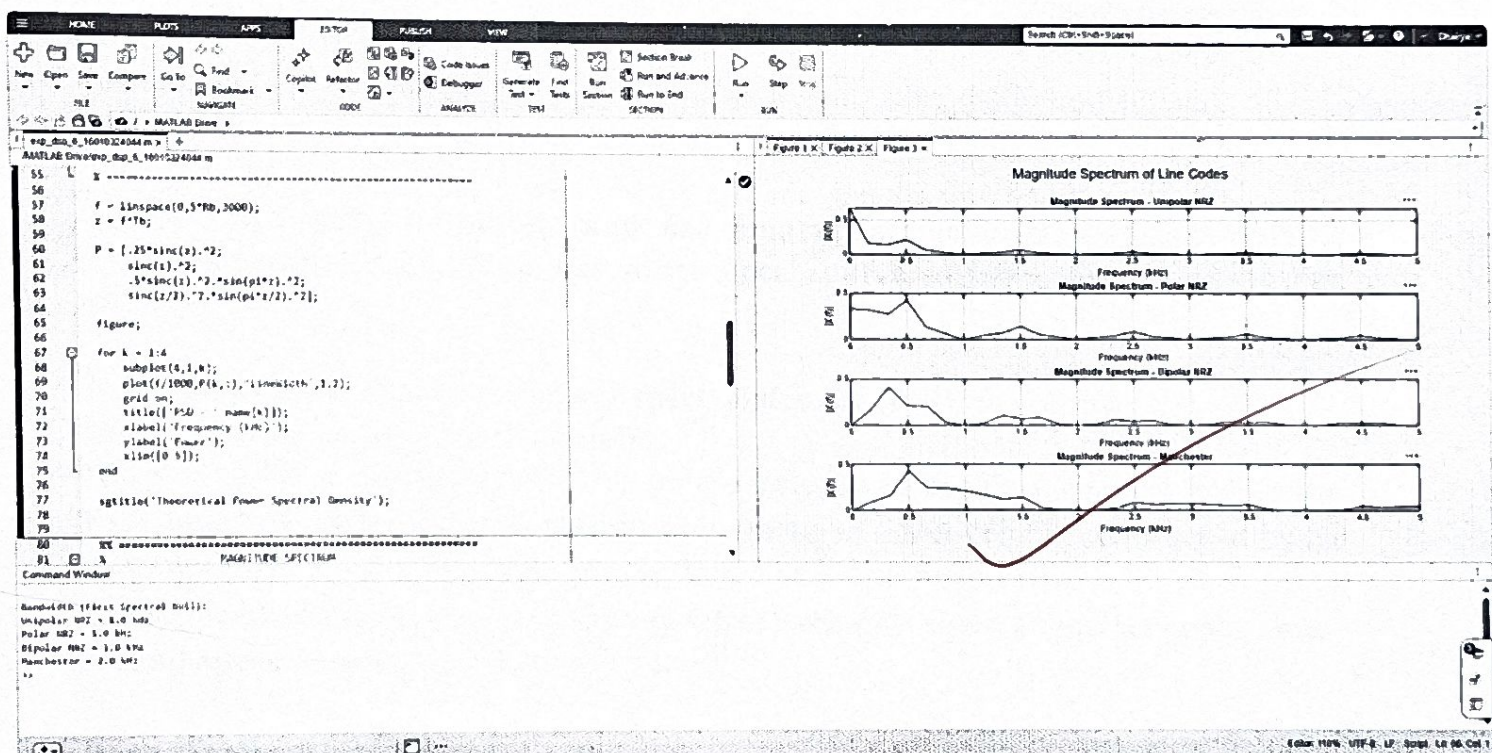
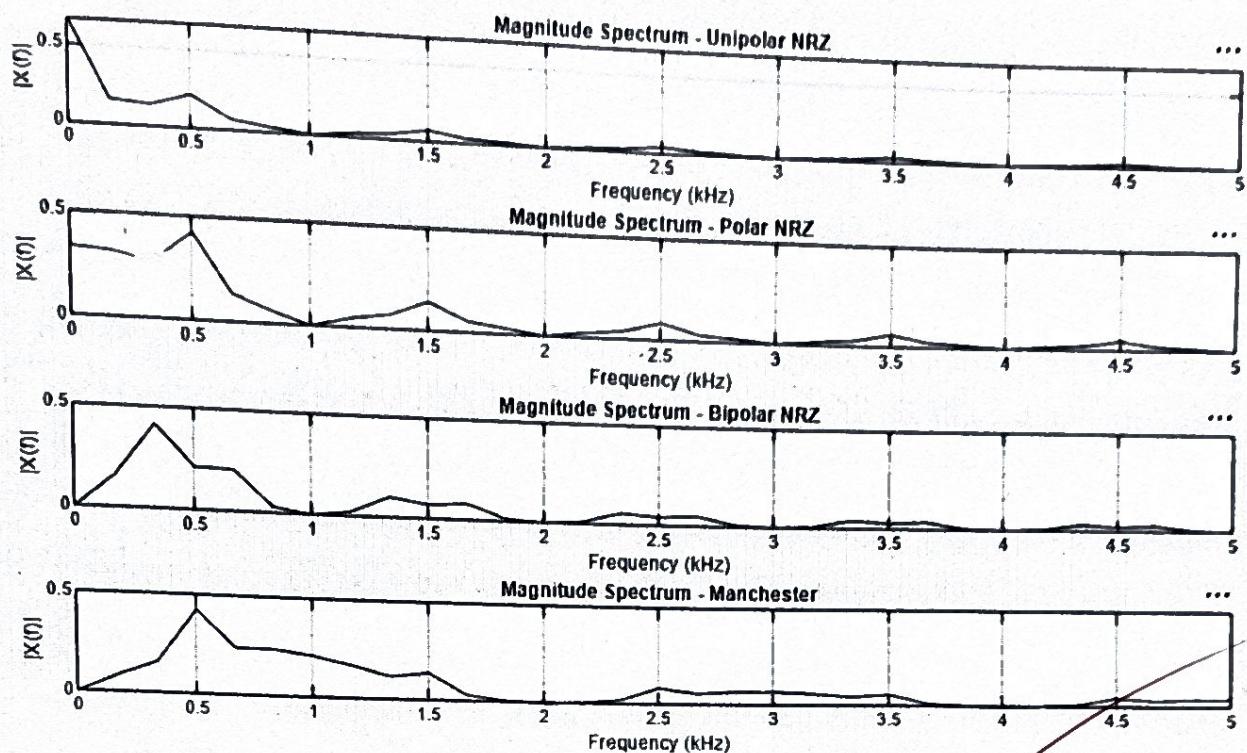


Figure 1 × Figure 2 × Figure 3 ×

Theoretical Power Spectral Density



Magnitude Spectrum of Line Codes



Discussion/Conclusion:

In this experiment, the waveforms of Unipolar NRZ, Polar NRZ, Bipolar NRZ and Manchester line coding were generated and their theoretical PSD and magnitude spectra were analyzed. It was observed that Manchester coding requires higher bandwidth (2 kHz) compared to NRZ schemes (1 kHz).

Signature of faculty in-charge